

CERD: Conditional Evidence Reconstruction and Decomposition for Incomplete Multimodal Learning

Shaowen Wan^a, Yanjun Lv^b, Lu Zhang^c, Dajiang Zhu^b, Bharat Biswal^a, Tianming Liu^d, Xiaobo Li^a and Lin Zhao^{a,*}

^aDepartment of Biomedical Engineering, New Jersey Institute of Technology, Newark, 07102, NJ, USA

^bDepartment of Computer Science and Engineering, The University of Texas at Arlington, Arlington, 76019, TX, USA

^cDepartment of Computer Science, Indiana University Indianapolis, Indianapolis, 46202, IN, USA

^dSchool of Computing, University of Georgia, Athens, 30602, GA, USA

ARTICLE INFO

Keywords:

Incomplete multimodal learning

Missing modality

Mixture-of-experts

Decision decomposition

Neurobiological prediction

ABSTRACT

Multimodal neurobiological prediction benefits from complementary imaging, genetic, clinical, and behavioral information, but real cohorts contain heterogeneous subject-specific missingness. Existing methods either operate only on the observed subset or reconstruct unavailable modalities, yet often do not distinguish generated representations from direct measurements when forming the final decision. We propose CERD, a Conditional Evidence Reconstruction and Decomposition framework for incomplete multimodal learning. CERD reconstructs missing-modality tokens from the same subject's observed modalities and uses provenance embeddings to preserve whether each representation is observed or generated. A Transformer with sparse top- k experts then models cross-modal interactions. Finally, global, modality-centered, and modality-pair-centered heads read the context-enriched slot features and produce complementary predictions. Their probabilities are combined according to learned input reliability, normalized-entropy confidence, and a branch prior. The stored branch probabilities and normalized weights exactly reproduce the neural prediction. Under arithmetic three-seed evaluation, CERD reaches 65.72% accuracy on three-class ADNI and 78.44% on family-disjoint ABCD current-ADHD classification. It leads all three ADNI metrics; among incomplete ABCD participants, it ties the highest accuracy and gives the highest macro-F1 and AUROC. Strict modality interventions yield model-output relevance that closely tracks accuracy-drop shares on both cohorts.

1. Introduction

Neurobiological phenotypes are studied using heterogeneous sources that describe different levels of the same subject, including brain structure and function, genetic susceptibility, cognitive and clinical measurements, and behavioral or environmental factors [12, 15, 35]. These sources are not interchangeable: each may provide diagnostic information that is absent from the others, while their agreement can strengthen a prediction [57, 38, 43, 22]. More broadly, multimodal learning exploits complementary views through representation, alignment, fusion, and co-learning strategies [1, 26, 30]. It therefore offers a useful basis for neurodevelopmental outcome prediction and neurodegenerative disease stratification, while its clinical use requires careful model development and evaluation [4]. Its practical value, however, depends on whether a model can use the evidence that is actually available for each subject rather than assuming a fixed, complete input set.

This assumption is frequently violated in research cohorts and clinical data. Modalities can be absent because of acquisition cost, protocol differences, contraindications, participant dropout, or failed quality control, producing many subject-specific availability patterns [50, 55, 11, 24, 56]. Restricting analysis to complete cases discards otherwise informative subjects, whereas replacing every missing modality

with a population-level constant removes the dependence between the unavailable source and the subject's observed evidence. Robust incomplete-modality learning must consequently address two coupled questions: how to infer a task-relevant representation for an unavailable modality, and how strongly that inferred representation should influence the final prediction. Existing aggregation and alignment methods avoid explicit completion but may leave recoverable cross-modal information unused [45, 21]. Generative approaches estimate missing observations or features, but many are specialized to commensurate imaging modalities or optimized primarily for reconstruction fidelity [49, 40, 3, 20, 59]. Flex-MoE supports arbitrary modality combinations through a missing-modality bank and sparse routing [54], yet its bank is indexed by the availability pattern and is therefore shared by subjects with the same observed subset. Less attention has been paid to jointly making completion subject-conditioned and making the downstream use of generated evidence reliability-aware.

The difficulty is not absence alone. Once a representation is inferred, the predictor must distinguish evidence supported by acquisition from evidence inferred through cross-modal association. Otherwise structural completion can be mistaken for evidential equivalence, allowing an uncertain reconstruction to influence fusion as if it were measured.

Missingness also changes what constitutes a useful explanation. A sparse router indicates where computation is allocated, but its gate value is not, by itself, the contribution

*Corresponding author.

 lin.zhao.1@njit.edu (L. Zhao)

ORCID(s):

of a modality to the prediction. Likewise, post-hoc attribution can audit a trained model, yet it need not reproduce the decision rule used during inference [23, 39, 31, 33]. For an incomplete subject, the most informative account distinguishes decision mass supported by observed representations from that supported by generated representations and shows how their certainty shapes the final output. This motivates a decision mechanism in which source status, predictive reliability, confidence, and branch contribution are represented explicitly and used in the prediction itself.

We propose CERD, a *Conditional Evidence Reconstruction and Decomposition* framework for incomplete multimodal learning. CERD maps heterogeneous modalities to a shared token space and reconstructs each unavailable target from a non-empty subset of the same subject's observed tokens. Provenance embeddings preserve whether a representation was observed or generated, and a Transformer with sparse top- k experts models interactions among the completed modality tokens. CERD then reads the fused state through global, modality-centered, and modality-pair-centered prediction heads and combines their probabilities using learned input reliability, normalized-entropy confidence, and a low-capacity prior. Because these branch predictions and weights are used directly in the forward pass, they provide an exact numerical decomposition of the neural probability.

We evaluate CERD on the Adolescent Brain Cognitive Development (ABCD) Study and the Alzheimer's Disease Neuroimaging Initiative (ADNI) [8, 47]. ABCD uses a strict binary current-ADHD endpoint, whereas ADNI distinguishes cognitively normal, mild-cognitive-impairment, and Alzheimer's-disease groups. Both use four heterogeneous modality groups and dataset-appropriate incomplete-modality protocols detailed in Section 4.1. Accuracy, macro-F1, and macro-AUROC are reported for both datasets. The principal contributions are summarized as follows:

- We propose subject-conditioned latent reconstruction, which infers each unavailable modality from the same subject's observed subset rather than assigning a surrogate determined only by the missingness pattern.
- We develop provenance-aware sparse-expert fusion by attaching observed/generated embeddings before self-attention and top- k routing, enabling modality interactions to remain conditioned on evidence origin.
- We formulate an exact multigranular decision decomposition over global, modality-centered, and modality-pair-centered branches. Predictive reliability, entropy-derived confidence, and a branch prior produce normalized weights whose probability mixture equals the neural output.
- Matched three-seed experiments on ADNI and a strict current-ADHD ABCD task evaluate predictive performance under heterogeneous missingness. Component controls isolate completion and sparse specialization,

while label-free decision changes and accuracy drops independently quantify modality dependence.

2. Related Work

2.1. Incomplete Multimodal Representation

Incomplete multimodal representation learning determines what information reaches a predictor when the observed subset varies across subjects [50, 55]. Subset-native methods directly encode the available inputs. HeMIS aggregates latent features with permutation-invariant statistics [11]; SMIL applies Bayesian meta-learning under severe missingness [24]; and M3Care models relations between patients and modalities in incomplete healthcare records [56]. Alignment methods instead make partial views compatible with complete ones. ShaSpec separates shared and source-specific factors [45], whereas MissModal combines geometric, distributional, and semantic constraints [21]. Missing-modality prompts adapt multimodal Transformers for visual recognition and affective analysis [16, 10]. SimMLM couples dynamic modality experts with an objective that relates predictions from richer and reduced views [19]. These methods preserve or align information from the current subset without constructing a target representation for every missing slot.

Conditional recovery explicitly supplies absent information. Multimodal variational autoencoders combine posteriors from individual modalities to infer a shared latent variable [49], and generalized multimodal evidence lower bounds extend training across observed subsets [40]. At the feature level, MMIN predicts unavailable input representations [58]; ActionMAE reconstructs randomly dropped sensing features [48]; and DiCMoR transfers observed distributions into a missing-modality space with normalizing flows [46]. Hetero-modal variational encoder-decoders jointly complete images and segment anatomy [3]. AGDiC recovers latent neuroimaging features with adaptive graphs and masked cross-attention [20], while ACADiff synthesizes missing MRI or PET volumes from images and metadata [59]. CERD targets heterogeneous sources. It generates one token slot per modality, supervised by held-out observed representations.

Pattern-conditioned completion occupies a useful middle ground. Flex-MoE learns a missing-modality bank indexed by the observed combination before sparse expert fusion [54]. CERD advances from pattern-level substitution to subject-level inference: each target generator queries the current subject's observed tokens, and provenance marks the resulting representation for downstream fusion and predictive-reliability estimation.

These families define different interfaces to the predictor under missingness. Subset-native and alignment models retain only measured information, whereas conditional recovery restores a consistently indexed modality layout. That fixed layout is particularly useful when downstream interactions refer to source identity. For heterogeneous sources,

recovery in a shared task space also avoids requiring imaging, genetic, clinical, and behavioral variables to share an observation space. CERD therefore reconstructs target-specific latent tokens rather than raw measurements and keeps their generated origin explicit.

This distinction also separates reconstruction fidelity from predictive utility. Matching the marginal geometry of a target modality does not ensure that a recovered representation carries class-discriminative evidence for a particular subject. CERD consequently places task-specific trust inside the prediction path, after subject-conditioned recovery, rather than treating distributional similarity as sufficient evidence.

2.2. Sparse Expert Fusion

Once a multimodal representation has been assembled, fusion determines how interaction capacity is allocated across its tokens. Transformer self-attention models cross-token context [42], whereas sparse mixtures of experts (MoE) condition the feed-forward transformation by routing each token to a small expert subset [37, 17, 5]. Routing designs control both specialization and expert utilization. BASE layers formulate balanced token assignment explicitly [18], expert-choice routing reverses the assignment direction so that experts select tokens [60], and MegaBlocks realizes dynamic expert computation with block-sparse operators [7].

Sparse experts have also been adapted to modality structure. V-MoE established scalable expert routing for visual tokens [32], and LIMoE showed that a shared expert pool can develop differentiated language–image processing [25]. MoE++ augments learned feed-forward experts with zero, copy, and constant transformations [13], while Flex-MoE specializes routing for arbitrary observed-modality combinations [54]. Together, these studies establish sparse routing as a mechanism for content-dependent representation specialization.

Incomplete inputs add a second routing signal beyond semantic content: whether a modality is measured, unavailable, or reconstructed. A fusion model can vary its token set with the observed subset, encode the availability pattern, or restore the missing positions before interaction. The last option preserves a stable source-indexed structure across subjects, but it also requires the router to recognize that measured and generated tokens carry different evidence status. CERD meets both requirements by routing a completed token layout after adding provenance, so expert specialization can depend jointly on content, source position, and observation status.

Interaction-structured experts address a complementary aspect of multimodal fusion. MMoE separates experts for unique, redundant, and synergistic information [53], and I²MoE adds weakly supervised interaction objectives and sample-specific reweighting [51]. CERD places sparse experts inside the provenance-aware completed-token backbone, so routing transforms observed and generated evidence in their shared context. A separate reliability–evidence

mechanism then determines how much each resulting decision view contributes to the prediction.

2.3. Interpretability and Decision Decomposition

Clinical and neurobiological models are often examined with post-hoc feature attribution, although clinical usefulness depends on whether the explanation is understandable and actionable for its intended user [41, 34, 2]. SHAP, integrated gradients, Grad-CAM, LIME, and counterfactual explanations estimate input relevance under different reference, gradient, localization, surrogate, or intervention constructions [23, 39, 36, 31, 44]. Multimodal variants shift the unit of analysis from individual features to information sources: MM-SHAP quantifies modality contributions and can expose single-modality dominance [28], while Layer-Wise Modality Decomposition traces modality-specific contributions through a pretrained fusion network [29]. These methods are valuable for examining arbitrary architectures, but the resulting attribution is external to the original prediction rule and may change with the explanation protocol.

Interaction-based interpretation asks whether modalities contribute unique, redundant, or synergistic information. MMoE and I²MoE encode these categories in specialized experts [53, 51], and LSMI estimates sample-level interactions using pointwise information-theoretic quantities [52]. Such analyses characterize relationships among modalities, but they do not automatically decompose the probability produced by a classifier. Under missingness, the origin and dependability of the evidence also become part of the decision. CERD addresses this requirement by predicting from several slot-centered views of one fused state and normalizing their contributions using learned modality reliability and predictive evidence. The resulting weights provide an exact numerical decomposition of the classifier probability and expose how the network combines its decision views.

Prior work therefore addresses representation recovery, sparse specialization, or attribution mostly as separate functions. CERD connects them in an ordered prediction path: subject-conditioned generation supplies missing latent evidence, provenance conditions token interaction and routing, and reliability controls its decision mass. This ordering prevents reconstruction from erasing source status and makes the final mixture an inspectable part of inference.

3. Methodology

3.1. Overview and Problem Formulation

CERD is an end-to-end framework for classifying subjects with incomplete multimodal data. Figure 1 presents the complete CERD pipeline from an incomplete input to its class probability, then divides it into three stages. First, conditional reconstruction infers unavailable tokens from the subject’s observed context (see Section 3.2). Second, provenance-aware sparse-expert fusion transforms the completed layout without erasing evidence origin (see Section 3.3). Finally, reliability-aware decomposition combines global, modality-centered, and pair-centered predictions

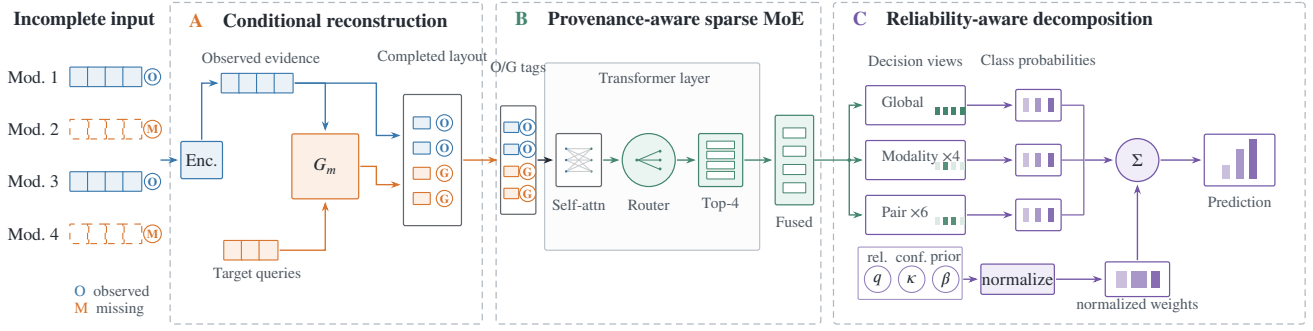


Figure 1: CERD reconstructs unavailable modality tokens from observed subject context, preserves observed/generated provenance through sparse expert fusion, and combines multigranular branch predictions using normalized reliability-aware weights.

(see Section 3.4). The sparse MoE is the shared backbone, and the normalized branch mixture is the prediction itself.

Throughout, n , m , and b index subjects, modalities, and decision branches, respectively; C is the number of classes. Let $\mathcal{M} = \{1, \dots, M\}$ be the modality set. The binary variable o_n^m records whether subject n 's input $\mathbf{x}_n^m$ is observed. A modality-specific encoder maps each observed input into P tokens of dimension D :

$$\mathbf{Z}_n^m = E_m(\mathbf{x}_n^m) \in \mathbb{R}^{P \times D}. \quad (1)$$

Let $\mathcal{O}_n = \{m : o_n^m = 1\}$ and $\mathcal{U}_n = \mathcal{M} \setminus \mathcal{O}_n$ denote the observed and missing sets. At least one modality must be observed, and every $m \in \mathcal{U}_n$ is generated from $\mathcal{O}_n$. CERD models $p(y_n | \{\mathbf{x}_n^m : m \in \mathcal{O}_n\}, \mathbf{o}_n)$, conditioning the prediction jointly on observed evidence and source availability. The availability vector determines provenance throughout the network, and generated evidence receives a learned predictive-reliability score below the observed reference value.

3.2. Conditional Evidence Reconstruction

A fixed missing-modality token assigns the same surrogate representation to every subject with the same availability pattern, so it cannot reflect differences in their observed evidence. CERD instead reconstructs each missing target conditionally from the current subject's available modalities. During training, a non-empty proper subset $\mathcal{S}_n \subsetneq \mathcal{O}_n$ is sampled as context and $\mathcal{T}_n = \mathcal{O}_n \setminus \mathcal{S}_n$ supplies reconstruction targets. Context modalities are independently dropped and the draw is repaired when necessary so that both $\mathcal{S}_n$ and $\mathcal{T}_n$ remain non-empty. This stochastic observed-subset construction exposes each generator to both rich and sparse contexts rather than only leave-one-modality-out cases.

For a reconstruction target $m \in \mathcal{T}_n$, we first collect the context tokens in $\mathbf{Z}_n^{\mathcal{S}}$. A target-specific generator then uses P learned queries $\mathbf{Q}_m$ to retrieve the evidence needed for modality m :

$$\begin{aligned} \mathbf{Z}_n^{\mathcal{S}} &= \text{Concat}_{j \in \mathcal{S}_n} \mathbf{Z}_n^j, \\ \hat{\mathbf{Z}}_n^m &= G_m(\mathbf{Q}_m, \mathbf{Z}_n^{\mathcal{S}}). \end{aligned} \quad (2)$$

The generator applies cross-attention followed by feed-forward refinement. Its refined query sequence $\mathbf{U}_n^m$ passes

through a channel gate:

$$\hat{\mathbf{Z}}_n^m = \mathbf{U}_n^m \odot \sigma(\mathbf{W}_m^g \mathbf{U}_n^m + \mathbf{b}_m^g). \quad (3)$$

The gate suppresses query responses unsupported by the current context. At inference, all observed modalities form $\mathbf{Z}_n^{\mathcal{S}}$, and generators are invoked only for missing targets. Thus, subjects with the same missing pattern can receive different reconstructions whenever their observed evidence differs.

Reconstruction is supervised at the pooled and token levels. Let $\bar{\mathbf{Z}}$ denote token averaging, R_m a target-specific projector, LN token-wise layer normalization, and sg stop-gradient. For target (n, m) ,

$$\begin{aligned} \ell_{\text{pool}}^{n,m} &= 1 - \cos\left(R_m(\bar{\mathbf{Z}}_n^m), \text{sg}[R_m(\bar{\mathbf{Z}}_n^m)]\right), \\ \ell_{\text{tok}}^{n,m} &= \text{SmoothL1}\left(\text{LN}(\hat{\mathbf{Z}}_n^m), \text{sg}[\text{LN}(\mathbf{Z}_n^m)]\right). \end{aligned} \quad (4)$$

For a mini-batch $\mathcal{N}$, these target losses are averaged as

$$\begin{aligned} N_{\text{rec}} &= \sum_{n \in \mathcal{N}} |\mathcal{T}_n|, \\ \mathcal{L}_{\text{rec}} &= \frac{1}{N_{\text{rec}}} \sum_{n \in \mathcal{N}} \sum_{m \in \mathcal{T}_n} \left(\ell_{\text{pool}}^{n,m} + \lambda_{\text{tok}} \ell_{\text{tok}}^{n,m} \right). \end{aligned} \quad (5)$$

The pooled term preserves subject-level semantic direction, whereas the token term preserves the ordered slot structure. Stop-gradient fixes the encoded target while the generator learns to approach it. One generator is reused for every context of target m : $\mathbf{Q}_m$ specifies the target, while $\mathbf{Z}_n^{\mathcal{S}}$ specifies the subject and available evidence. CERD therefore avoids a separate generator for every missingness pattern.

The two reconstruction levels constrain complementary failure modes. Pooled alignment alone can preserve a subject-level mean while losing slot organization, whereas token alignment alone can reproduce local structure without retaining the target's semantic direction. Their joint use makes generated tokens compatible with the fixed fusion layout at both resolutions.

3.3. Provenance-Aware Sparse MoE Fusion

Reconstruction supplies a complete latent layout, but reconstructed information should not be indistinguishable

from an observed measurement. Before fusion, CERD therefore adds an observed or generated provenance embedding:

$$\tilde{\mathbf{Z}}_n^m = \begin{cases} \mathbf{Z}_n^m + \mathbf{e}^{\text{obs}}, & o_n^m = 1, \\ \hat{\mathbf{Z}}_n^m + \mathbf{e}^{\text{gen}}, & o_n^m = 0. \end{cases} \quad (6)$$

These embeddings retain token origin while keeping all modalities in the same latent space.

The marked tokens are concatenated and processed by global self-attention. The Transformer feed-forward block is replaced by a sparse MoE layer. For token $\mathbf{h}$, router $\mathbf{g}$ defines the active expert set $\mathcal{A}(\mathbf{h})$, after which only those expert outputs are combined:

$$\begin{aligned} \mathcal{A}(\mathbf{h}) &= \text{TopK}(\mathbf{g}(\mathbf{h}), k), \\ \text{MoE}(\mathbf{h}) &= \sum_{e \in \mathcal{A}(\mathbf{h})} \pi_e(\mathbf{h}) F_e(\mathbf{h}). \end{aligned} \quad (7)$$

Here, F_e is expert e , and π_e is its softmax weight within $\mathcal{A}(\mathbf{h})$. Because provenance is added before routing, expert selection can respond to modality content and observed-generated status. A load-balancing term promotes utilization of the expert pool. The fused sequence is finally separated by modality and pooled into context-enriched features $\{\mathbf{f}_n^m\}_{m=1}^M$.

The attention and expert stages have complementary roles. Global self-attention communicates evidence across every modality position, whereas sparse feed-forward experts specialize the transformation applied to each resulting token. The router therefore adapts computation to both the subject and the observed-generated composition without creating a separate network for every missingness pattern. With $M = 4$ and $P = 16$, the shared sequence contains 64 tokens. ADNI activates four of 16 experts per token, while ABCD activates two of eight. The MoE remains a shared representation backbone, and all subsequent diagnostic branches reuse its pooled modality features.

Routing follows self-attention, so expert assignments are computed from tokens that already encode whole-subject context rather than isolated unimodal features. Sparse specialization therefore refines cross-modal evidence while preserving one common modality layout; it does not partition the sources into independent processing streams.

3.4. Reliability-Aware Evidence Decomposition

A single readout compresses the fused state into one decision view. CERD instead uses three readout granularities. Let 0 denote the global branch, $m \in \mathcal{M}$ a modality-centered branch, and $(i, j) \in \mathcal{P} = \{(i, j) : i < j\}$ a pair-centered branch. The complete branch set is $\mathcal{B} = \{0\} \cup \mathcal{M} \cup \mathcal{P}$, and every branch produces a class-probability vector $\mathbf{p}_n^b$. Because the readouts follow global attention and sparse experts, they are anchored to particular modality positions but retain cross-modal context. A pair branch receives

$$\mathbf{u}_n^{(i,j)} = [\mathbf{f}_n^i; \mathbf{f}_n^j; \mathbf{f}_n^i \odot \mathbf{f}_n^j; |\mathbf{f}_n^i - \mathbf{f}_n^j|], \quad (8)$$

which represents agreement and discrepancy between its two anchors. With four modalities, $\mathcal{B}$ contains one global, four modality-centered, and six pair-centered heads.

CERD then separates learned input reliability from predictive confidence. Let a_m be a small modality-specific reliability head. Because every missing target is generated from at least one observed source, modality reliability has two cases:

$$r_n^m = \begin{cases} 1, & o_n^m = 1, \\ \sigma(a_m(\mathbf{f}_n^m)), & o_n^m = 0. \end{cases} \quad (9)$$

Observed evidence establishes the reference value, while generated evidence receives a subject-dependent score in $(0, 1)$. Since r_n^m is optimized through the prediction objective, it quantifies task-specific predictive trust. The raw scores are estimated independently; normalization is applied when branches compete for decision weight and when cohort-level shares are visualized.

Branch reliability is then obtained directly from the modalities anchoring each readout:

$$q_n^b = \begin{cases} \frac{1}{M} \sum_{m=1}^M r_n^m, & b = 0, \\ r_n^m, & b = m, \\ \sqrt{r_n^i r_n^j}, & b = (i, j). \end{cases} \quad (10)$$

The global mean remains on the scale of one reliability score, while the geometric mean requires support from both pair anchors. Predictive confidence is measured independently from the entropy H of the C -class branch distribution:

$$\kappa_n^b = \max \left(1 - \frac{H(\text{sg}[\mathbf{p}_n^b])}{\log C}, \epsilon \right). \quad (11)$$

This score measures distributional concentration and is used to compare decision views [9, 14, 27].

Because every modality position is either observed or conditionally generated, all decision views are evaluated. Original availability enters through provenance and Eq. (9). A learned scalar β_b supplies a prior for branch b . The unnormalized evidence score, normalized branch weight, and final prediction are

$$\begin{aligned} s_n^b &= (q_n^b + \epsilon)(\kappa_n^b + \epsilon)e^{\beta_b}, \\ w_n^b &= \frac{s_n^b}{\sum_{b' \in \mathcal{B}} s_n^{b'}}, \\ \mathbf{p}_n &= \sum_{b \in \mathcal{B}} w_n^b \mathbf{p}_n^b. \end{aligned} \quad (12)$$

Thus, reliability q_n^b , confidence κ_n^b , and prior β_b have separate roles before normalization. The stored $\mathbf{p}_n^b$ and w_n^b are forward-pass quantities, and their weighted sum exactly reproduces $\mathbf{p}_n$.

Reliability and confidence serve distinct roles. A generated branch may be sharply peaked yet weakly supported by its inferred input; an observed view may be reliable but class-ambiguous. Their product requires both source support and discriminative concentration, while the branch prior

captures stable utility without replacing subject-dependent variation.

For visualization, branch weights are summarized at the modality level. A modality-centered weight is assigned to its anchor, a modality-pair-centered weight is divided equally between its two anchors, and the global weight is distributed uniformly:

$$d_n^m = w_n^m + \frac{1}{2} \sum_{j \neq m} w_n^{(m,j)} + \frac{w_n^0}{M}. \quad (13)$$

The resulting modality decision allocation is a descriptive summary of the model decision, rather than an additional prediction mechanism.

3.5. Training Objective

CERD is trained end to end with five losses, each attached to a defined part of the method chain:

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_b \mathcal{L}_{\text{branch}} + \lambda_r \mathcal{L}_{\text{rec}} + \lambda_g \mathcal{L}_{\text{balance}} + \mathcal{L}_{\text{rob}}. \quad (14)$$

Here, $\mathcal{L}_{\text{cls}} = |\mathcal{N}|^{-1} \sum_{n \in \mathcal{N}} \text{CE}(y_n, \mathbf{p}_n)$ supervises the fused probability, and $\mathcal{L}_{\text{rec}}$ is defined in Eq. (5). Each branch is also trained directly. We first normalize its stopped reliability within mini-batch $\mathcal{N}$, then apply reliability-weighted cross-entropy:

$$\begin{aligned} \alpha_n^b &= \frac{\text{sg}[q_n^b]}{\sum_{u \in \mathcal{N}} \text{sg}[q_u^b] + \epsilon}, \\ \mathcal{L}_{\text{branch}} &= \frac{1}{|\mathcal{B}|} \sum_{b \in \mathcal{B}} \sum_{n \in \mathcal{N}} \alpha_n^b \text{CE}(y_n, \mathbf{p}_n^b). \end{aligned} \quad (15)$$

This term supervises every readout independently of its current mixture weight. For sparse layer ℓ , let $\mathbf{I}_\ell$ collect the router probability received by each expert and $\mathbf{L}_\ell$ its assigned-token counts. The load loss $\mathcal{L}_{\text{balance}} = \sum_\ell [\text{CV}^2(\mathbf{I}_\ell) + \text{CV}^2(\mathbf{L}_\ell)]$ promotes balanced expert use.

Robustness is trained from a strict non-empty reduced view $\tilde{\mathcal{O}}_n \subset \mathcal{O}_n$. Let $\mathbf{p}_n$ and $\tilde{\mathbf{p}}_n$ be the original- and reduced-view predictions. The batch-averaged robustness loss is

$$\begin{aligned} \mathcal{L}_{\text{rob}} &= \frac{1}{|\mathcal{N}|} \sum_{n \in \mathcal{N}} \left[\lambda_d \text{CE}(y_n, \tilde{\mathbf{p}}_n) \right. \\ &\quad + \lambda_{\text{sd}} T^2 \text{KL}(\text{sg}[\mathbf{p}_n^{(T)}] \parallel \tilde{\mathbf{p}}_n^{(T)}) \\ &\quad \left. + \lambda_{\text{mf}} [\text{CE}(y_n, \mathbf{p}_n) - \text{CE}(y_n, \tilde{\mathbf{p}}_n)]_+ \right]. \end{aligned} \quad (16)$$

Superscript (T) denotes temperature-scaled probabilities. The three lines respectively train reduced-view classification, transfer the original-view class distribution, and preserve the advantage of richer evidence. Reduced-view training constrains the final decision across availability patterns, whereas stochastic reconstruction contexts provide target supervision to the generators.

These two subset constructions impose different invariances. Reconstruction holds out observed targets to learn cross-modal recovery, whereas reduced-view training perturbs availability at the prediction level and constrains the

resulting class distribution. Their combination couples target recovery to stable decisions rather than optimizing reconstruction and robustness as separate objectives.

3.6. Inference and Computational Structure

At inference, CERD first encodes the observed sources and conditionally generates only the missing targets. All modality positions then enter the provenance-aware backbone in a fixed MP -token layout. Sparse routing activates k feed-forward experts per token, while the attention path and parameters remain shared across availability patterns. The branch heads add only $1 + M + \binom{M}{2}$ shallow classifiers. Equation (12) then returns both the class probability and the exact weights that produced it. No reconstruction loss, reduced-view pass, or auxiliary explanation model is required during inference.

This organization has two practical consequences. First, the number of possible missingness patterns grows exponentially with M , but CERD does not store a pattern-specific completion vector or prediction network. Target-specific generators are reused with any non-empty context, and sparse experts specialize at the token level. Second, the decision trace is available in the ordinary forward output: modality reliability describes the status of each supplied representation, branch probabilities describe the decisions supported at different granularities, and normalized weights state how those decisions were combined.

4. Experiments

4.1. Datasets and Preprocessing

ABCD. The ABCD Study characterizes adolescent brain, cognitive, behavioral, and health development [8]. We derive a strict current-ADHD case-control endpoint from the baseline parent K-SADS assessment. Controls have no current, past, partial-remission, or unspecified ADHD and at most three current symptoms in either nine-item domain. Cases require a current full ADHD diagnosis, explicit 0/1 coding for all 18 current symptoms, and at least six symptoms in one domain. Past-only, remission-only, unspecified, and symptom-only participants are excluded rather than assigned to either class. From 848 eligible cases, 1,272 controls are selected by a fixed label-blind order before family grouping. The resulting 2,120-participant cohort is split family-disjointly into 1,520/297/303 for training/validation/testing, with control/case counts 917/603, 177/120, and 178/125.

ABCD predictors form four model modalities. Imaging (785 variables) concatenates rs-fMRI network connectivity and regional variance, regional beta contrasts from SST, n-back, and MID task-fMRI, regional T1 gray-matter volume, and DTI-derived FA. Genetics comprises 1,150 directly encoded SNP dosages retained after training-set missingness, minor-allele-frequency, and LD pruning within 14 prespecified candidate-gene windows; no PCA, PRS, or ancestry components are used. The cognition and health group contains 90 NIH Toolbox, WISC, Little Man, RAVLT, sleep, and physical-activity variables. The behavior and environment group begins with 221 non-ADHD behavioral, family,

neighborhood, socioeconomic, and address-linked exposure variables, of which 219 survive training-only screening.

ADNI. The Alzheimer’s Disease Neuroimaging Initiative (ADNI) is a multicenter longitudinal study of biomarkers for cognitive decline and Alzheimer’s disease [47]. We classify participants as cognitively normal, mild cognitive impairment, or Alzheimer’s disease. Its four modalities are 187 regional FreeSurfer structural-MRI measurements, 387,639 directly encoded SNPs, 4,189 clinical and cognitive variables, and 151 biospecimen measurements. The cohort contains 2,116 participants: 1,480 for training, 318 for validation, and 318 for testing. Their class counts are 648/522/310, 139/112/67, and 139/112/67, respectively. In both datasets, a subject is retained when at least one selected modality is available.

Missing-modality patterns. ADNI follows source-table availability: 27.64%, 28.62%, and 31.13% of the training, validation, and test participants are incomplete, respectively. Every incomplete ADNI participant lacks exactly one of the four modalities, and none is all-missing. For ABCD, a fixed label-independent mask augments source-table availability to approximately 15% incomplete participants per split. Among additionally masked complete cases, one, two, or three modalities are withheld in proportions 0.72, 0.23, and 0.05. The all-missing case is forbidden, and every method receives the same split-specific inputs. Table 1 gives the exact final counts.

Table 1: Incomplete participants in the fixed splits; entries are n/N (%).

Split	ADNI, n/N (%)	ABCD, n/N (%)
Training	409/1480 (27.64)	228/1520 (15.00)
Validation	91/318 (28.62)	45/297 (15.15)
Testing	99/318 (31.13)	45/303 (14.85)

Preprocessing. For ABCD, imaging and tabulated variables with excessive missingness or near-zero variance are removed, after which available entries are median-imputed and standardized using training statistics. Genetic selection is likewise frozen from training participants: SNPs exceeding 2% missingness or below 1% minor-allele frequency are removed, followed by LD pruning with a 50-SNP window, five-SNP step, and $r^2 = 0.8$ threshold. The retained 0/1/2 dosages are median-imputed and standardized without PCA. For ADNI, modality tables are joined by PTID and repeated FreeSurfer exports are deduplicated by database update stamp. Regional structural measurements are mean-imputed and standardized, genotypes are mode-imputed and transformed to $[-1, 1]$, and missing entries in clinical and biospecimen tables are filled by training-column means. In both cohorts, entry-level imputation is applied only within an available modality. A wholly unavailable modality remains masked and is reconstructed only by CERD. All preprocessing statistics and feature-selection decisions are estimated from the training split.

4.2. Experimental Setup

Implementation details.

CERD uses $P = 16$ tokens per modality, hidden dimension $D = 128$, and one four-head fusion layer. ADNI uses 16 experts with top-4 routing; ABCD uses eight experts with top-2 routing. Each target generator contains two cross-attention layers. We use $(\lambda_b, \lambda_g) = (0.1, 0.01)$, $(\lambda_d, \lambda_{sd}, \lambda_{mf}) = (0.1, 0.15, 0.1)$, $T = 2$, and $\epsilon = 10^{-3}$. The reconstruction coefficient is 1.0 on ADNI and 0.25 on ABCD. Models are optimized for at most 50 epochs with AdamW; learning rates are 1.25×10^{-4} and 1.0×10^{-4} , respectively. ADNI checkpoints are selected by validation macro-F1. For binary ABCD, positive-class AUPRC selects the checkpoint and a validation-only threshold maximizes macro-F1; that threshold is frozen before test inference. Results are arithmetic mean $\pm$ sample standard deviation over seeds 0/1/2 for ADNI and 31/32/33 for ABCD; probabilities are never ensembled across seeds.

Evaluation metrics. We report accuracy, macro-F1, and macro-AUROC. Accuracy measures overall decisions, macro-F1 weights every endpoint class equally, and macro-AUROC summarizes class ranking independently of one operating point. The experiments ask whether CERD improves prediction under heterogeneous missingness, whether each stage contributes, whether performance is retained for incomplete participants, and whether model-output relevance agrees with controlled modality interventions. ABCD comparisons use the same family-disjoint split and participant-level mask for every method, with training-split preprocessing throughout.

The incomplete stratum is reported separately because aggregate performance can conceal a model that succeeds mainly on complete participants. Strict removal then holds participant identity fixed and turns the descriptive allocation into a testable prediction: removing evidence assigned greater decision mass should produce the larger score change.

4.3. Comparison with Baselines

Compared methods. We compare CERD with six representative methods for incomplete multimodal learning. Flex-MoE combines a missing-modality bank with sparse routing [54]. I²MoE models unique, redundant, and synergistic multimodal interactions [51], whereas MoE++-corrected introduces learned, zero, copy, and constant experts [13]. AnyMod uses modality queries and a task anchor to accommodate arbitrary input combinations [6]. We further include task-matched implementations inspired by AGDiC and ACADiff because their original formulations operate on different neuroimaging representations [20, 59]. All methods use the same participant splits, modality groups, missingness patterns, and evaluation metrics.

Main results. Table 2 shows that CERD ranks first on all three ADNI metrics. Relative to the strongest competing entry in each column, its mean accuracy, macro-F1, and AUROC are higher by 0.94, 1.01, and 1.30 points, respectively. The concurrent gains in macro-F1 and AUROC indicate that

Table 2

Comparison on ADNI (%). AUC denotes macro-AUROC; the best values are boldfaced.

Method	Acc.	Macro-F1	AUC
Flex-MoE [54]	62.26±1.09	60.62±2.12	77.75±1.03
I ² MoE [51]	64.05±2.27	61.83±1.89	79.35±1.63
MoE++-corrected [13]	59.75±0.54	58.85±2.06	79.23±1.39
AnyMod [6]	64.78±4.63	63.55±4.28	79.77±2.12
AGDiC-inspired [20]	58.60±3.36	56.54±2.38	75.10±1.36
ACADiff-inspired [59]	55.87±0.48	52.14±1.20	71.10±1.24
CERD (Ours)	65.72±1.09	64.56±2.09	81.07±0.55

Table 3

Comparison on family-disjoint ABCD current-ADHD classification with approximately 15% incomplete participants (%). AUC denotes macro-AUROC; best means are boldfaced.

Method	Acc.	Macro-F1	AUC
Flex-MoE	78.22±0.33	76.56±0.67	86.23±0.33
I ² MoE	78.33±1.01	76.63±1.28	86.08±0.25
MoE++-corrected	77.12±0.69	76.34±0.47	85.78±0.21
AnyMod	79.43±1.66	78.51±1.26	84.07±1.62
AGDiC-inspired	80.31±1.01	79.21±0.69	85.38±0.52
ACADiff-inspired	78.11±2.10	76.80±1.94	82.59±1.44
CERD (Ours)	78.44±0.76	77.81±0.35	85.12±0.05

Table 4

CERD performance across test availability strata (% , mean ± SD).

Data	Input	<i>n</i>	Acc.	Macro-F1	AUC
ADNI	Complete	219	65.45±1.47	64.65±1.44	80.99±0.87
	Incomplete	99	66.33±2.92	62.86±4.08	82.35±1.30
ABCD	Complete	258	78.42±0.22	77.94±0.27	85.45±0.33
	Incomplete	45	78.52±4.63	76.70±3.52	85.70±1.30

the benefit is not confined to the majority diagnosis: CERD improves class-balanced decisions while preserving stronger one-vs-rest ranking.

On the complete-plus-incomplete ABCD cohort, Table 3 shows closely grouped discriminative performance rather than a universal winner: AGDiC-inspired has the highest accuracy and macro-F1, whereas Flex-MoE has the highest AUROC. CERD remains within 1.87 points of the best mean in every column and has the smallest seed variation in macro-F1 and AUROC. Its missing-input behavior is more distinctive. In the 45-participant incomplete stratum, CERD reaches 78.52% accuracy, 76.70% macro-F1, and 85.70% AUROC; it ties the best accuracy and exceeds the next-best macro-F1 and AUROC by 1.05 and 1.18 points, respectively. Thus its main advantage is concentrated where conditional evidence recovery is active.

Table 4 shows that CERD remains operational for incomplete participants in both cohorts. ABCD accuracy is essentially unchanged across availability strata, while macro-F1 decreases by 1.24 points; the larger uncertainty reflects the smaller 45-participant incomplete group. Because the

strata contain different participants, they describe deployment conditions rather than a causal effect of missingness; the matched within-participant removals in Section 4.5 isolate modality dependence directly.

4.4. Ablation Study

Compared variants. Each control starts from the dataset's full configuration and removes one named mechanism. Conditional completion is bypassed; provenance embeddings are suppressed; the sparse expert feed-forward block is replaced by an aligned dense FFN; multigranular decomposition is reduced to the global head and its associated specialist losses are removed; or adaptive branch scores are replaced by a uniform simplex. Every variant is independently trained and validation-selected with the same three seeds as its main result.

Ablation results. Table 5 identifies conditional completion and multigranular readout as the largest cross-cohort effects. Their removal lowers ADNI accuracy by 1.99 and 3.25 points and incomplete-ABCD macro-F1 by 5.18 and 3.91 points, respectively. Sparse specialization is specifically useful when evidence is unavailable: relative to the aligned Dense FFN, CERD gains 2.22 accuracy, 3.66 macro-F1, and 1.33 AUROC points on incomplete ABCD inputs. On the full ABCD test cohort, Dense is 0.66 points higher in accuracy, showing that the sparse gain is concentrated in the target missing-input regime rather than complete cases. Provenance produces a smaller isolated change, while learned reliability principally improves hard decisions over uniform weighting.

4.5. Modality Contribution Analysis

We characterize source use by strict leave-one-modality-out interventions on originally complete test participants. The selected input block is zeroed, marked unavailable, and excluded from conditional completion, holding participant identity and the trained model fixed. Model-output relevance is the rate at which this intervention changes the predicted class, normalized across the four modalities within each seed. Independently, positive accuracy decreases are normalized in the same way. Each four-entry vector therefore sums to 100%, and their correspondence tests whether label-free prediction sensitivity recovers the empirical performance ordering. All seed-level quantities are computed before arithmetic aggregation.

Table 6 and Fig. 2 show that the label-free model response reproduces the intervention ordering. Pearson correlations between relevance and drop share are 0.979 on ADNI and 0.947 on ABCD; ABCD also has identical rank order (Spearman $\rho = 1$). ADNI concentrates dependence in structural MRI and biospecimens, placing molecular and anatomical evidence above the high-dimensional clinical table. ABCD instead increases monotonically from imaging through genetics and cognition–health to behavior–environment. The last source changes the most predictions and produces a 21.06-point accuracy decrease, consistent with a current behavioral diagnosis while retaining nonzero neuroimaging and SNP dependence.

Table 5

Matched component analysis (% , mean $\pm$ SD). ADNI columns use its full test cohort; ABCD columns use the originally incomplete test stratum ($n = 45$), where completion and provenance are active.

Configuration	ADNI, all			ABCD, incomplete		
	Acc.	Macro-F1	AUC	Acc.	Macro-F1	AUC
w/o Conditional Completion	63.73 $\pm$ 0.79	62.82 $\pm$ 1.58	79.48 $\pm$ 0.41	74.07 $\pm$ 6.42	71.52 $\pm$ 4.38	82.67 $\pm$ 2.78
w/o Provenance Embeddings	63.52 $\pm$ 1.57	63.41 $\pm$ 1.95	80.50 $\pm$ 0.70	79.26 $\pm$ 1.28	76.55 $\pm$ 2.25	85.48 $\pm$ 0.93
w/o Sparse MoE (Dense FFN)	64.99 $\pm$ 1.19	65.27 $\pm$ 1.12	81.59 $\pm$ 0.74	76.30 $\pm$ 3.39	73.04 $\pm$ 3.66	84.37 $\pm$ 1.22
w/o Multigranular Decomposition	62.47 $\pm$ 0.79	62.36 $\pm$ 1.00	79.72 $\pm$ 1.32	74.81 $\pm$ 5.59	72.79 $\pm$ 3.92	84.74 $\pm$ 1.48
w/o Reliability-aware Weights	64.05 $\pm$ 1.49	63.88 $\pm$ 0.21	80.95 $\pm$ 0.19	74.07 $\pm$ 8.41	71.50 $\pm$ 6.46	85.70 $\pm$ 0.34
Full CERD	65.72$\pm$1.09	64.56$\pm$2.09	81.07$\pm$0.55	78.52$\pm$4.63	76.70$\pm$3.52	85.70$\pm$1.30

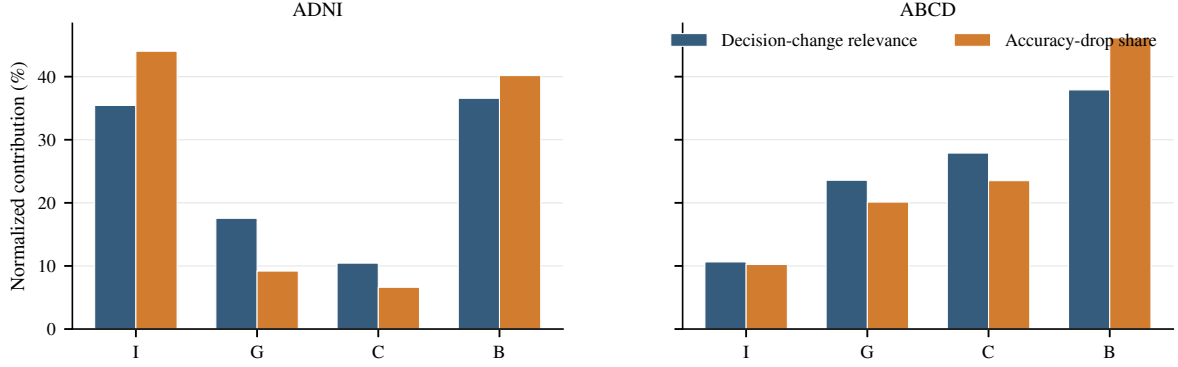


Figure 2: Normalized model-output relevance and accuracy-drop shares under strict modality removal (three-seed means). I/G/C/B denote MRI/genetics/clinical/biospecimen on ADNI and imaging/genetics/cognition–health/behavior–environment on ABCD.

Table 6

Strict-removal modality audit (mean $\pm$ SD). Rel. is normalized prediction-change relevance; Drop is normalized positive accuracy decrease; Δ Acc. is the unnormalized decrease in percentage points.

Data	Modality	Rel. (%)	Drop (%)	Δ Acc.
ADNI	MRI	35.44 $\pm$ 3.79	44.03 $\pm$ 3.10	43.07 $\pm$ 1.47
	Genetics	17.55 $\pm$ 3.82	9.19 $\pm$ 10.52	9.28 $\pm$ 12.36
	Clinical	10.44 $\pm$ 2.25	6.61 $\pm$ 3.80	6.70 $\pm$ 4.34
	Biospecimen	36.57 $\pm$ 1.90	40.17 $\pm$ 9.75	38.81 $\pm$ 5.94
ABCD	Imaging	10.63 $\pm$ 7.55	10.23 $\pm$ 17.72	2.58 $\pm$ 8.17
	Genetics	23.58 $\pm$ 13.37	20.12 $\pm$ 19.76	9.04 $\pm$ 10.45
	Cog./health	27.89 $\pm$ 6.63	23.51 $\pm$ 13.91	11.50 $\pm$ 8.51
	Behav./env.	37.90 $\pm$ 4.79	46.14 $\pm$ 6.98	21.06 $\pm$ 0.22

5. Discussion

Coupling completion with controlled trust. CERD leads all three ADNI metrics and is strongest on macro-F1 and AUROC in the incomplete ABCD stratum. Removing conditional completion lowers every metric in both settings, linking subject-conditioned recovery to the regime in which unavailable evidence must actually be represented.

Sparse specialization of completed evidence. Provenance precedes routing, so experts specialize using semantic content, modality position, and evidence origin. The aligned Dense-FFN control retains reconstruction and decomposition; its full-cohort advantage reverses on incomplete ABCD, where sparse routing gains 2.22 accuracy and 3.66 macro-F1 points. The gain thus arises from heterogeneous

observed–generated token layouts rather than complete-case capacity.

Why multigranular decomposition matters. Reducing CERD to its global head produces the largest ADNI accuracy loss and a 3.91-point incomplete-ABCD macro-F1 loss. Because each $\mathbf{f}_n^m$ already contains cross-modal context, modality and pair anchors expose source-centered and agreement–discrepancy views of a shared state, adding decision structure beyond one pooled representation.

Prediction consistency. ADNI depends most on structural MRI and biospecimens, whereas current-ADHD decisions are most sensitive to behavior/environment and cognition/health. Prediction-change relevance tracks independently measured accuracy-drop shares in both cohorts, turning source ordering into a direct faithfulness result for the frozen classifier.

6. Conclusion

CERD combines conditional token recovery, sparse provenance-aware expert fusion, and reliability-based evidence decomposition. It gives the strongest result across all ADNI metrics and the strongest class-balanced discrimination among incomplete ABCD participants. Across cohorts, controls isolate completion, multigranular readout, and missing-input expert specialization. Normalized decision changes agree with strict-removal accuracy effects, connecting the model’s source sensitivity to measured predictive dependence.

References

- [1] Baltrusaitis, T., Ahuja, C., Morency, L.P., 2019. Multimodal machine learning: A survey and taxonomy. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 41, 423–443. doi:10.1109/TPAMI.2018.2798607.
- [2] Borys, K., Schmitt, Y.A., Nauta, M., Seifert, C., Krämer, N., Friedrich, C.M., Nensa, F., 2023. Explainable ai in medical imaging: An overview for clinical practitioners - saliency-based xai approaches. *European Journal of Radiology*.
- [3] Dorent, R., Joutard, S., Modat, M., Ourselin, S., Vercauteren, T., 2019. Hetero-modal variational encoder-decoder for joint modality completion and segmentation, in: *Medical Image Computing and Computer Assisted Intervention – MICCAI 2019*, Springer. pp. 74–82. doi:10.1007/978-3-030-32245-8_9.
- [4] Esteva, A., Robicquet, A., Ramsundar, B., Kuleshov, V., DePristo, M., Chou, K., Cui, C., Corrado, G., Thrun, S., Dean, J., 2019. A guide to deep learning in healthcare. *Nature Medicine*.
- [5] Fedus, W., Zoph, B., Shazeer, N., 2022. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. *Journal of Machine Learning Research* 23, 1–39.
- [6] Feng, Y., Gao, B., Deng, S., Qiu, A., Qin, J., 2024. Unified multi-modal learning for any modality combinations in alzheimer’s disease diagnosis, in: *Medical Image Computing and Computer Assisted Intervention – MICCAI 2024*, Springer Nature Switzerland. pp. 487–497. doi:10.1007/978-3-031-72384-1_46.
- [7] Gale, T., Narayanan, D., Young, C., Zaharia, M., 2023. MegaBlocks: Efficient sparse training with mixture-of-experts. *Proceedings of Machine Learning and Systems* 5.
- [8] Garavan, H., Bartsch, H., Conway, K., Decastro, A., Goldstein, R.Z., Heeringa, S., Jernigan, T., Potter, A., Thompson, W., Zahs, D., 2018. Recruiting the ABCD sample: Design considerations and procedures. *Developmental Cognitive Neuroscience* 32, 16–22. doi:10.1016/j.dcn.2018.04.004.
- [9] Guo, C., Pleiss, G., Sun, Y., Weinberger, K.Q., 2017. On calibration of modern neural networks, in: *Proceedings of the 34th International Conference on Machine Learning*, PMLR. pp. 1321–1330.
- [10] Guo, Z., Jin, T., Zhao, Z., 2024. Multimodal prompt learning with missing modalities for sentiment analysis and emotion recognition, in: *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, Association for Computational Linguistics. pp. 1726–1736. doi:10.18653/v1/2024.acl-long.94.
- [11] Havaei, M., Guizard, N., Larochelle, H., Jodoin, P., 2016. Hemis: Hetero-modal image segmentation. *Medical Image Computing and Computer-Assisted Intervention*.
- [12] Jack, C.R., Bennett, D.A., Blennow, K., Carrillo, M.C., Dunn, B., Haeberlein, S.B., Holtzman, D.M., Jagust, W., Jessen, F., Karlawish, J., et al., 2018. NIA-AA research framework: Toward a biological definition of alzheimer’s disease. *Alzheimer’s & Dementia*.
- [13] Jin, P., Zhu, B., Yuan, L., Yan, S., 2025. MoE++: Accelerating mixture-of-experts methods with zero-computation experts, in: *International Conference on Learning Representations*.
- [14] Lakshminarayanan, B., Pritzel, A., Blundell, C., 2017. Simple and scalable predictive uncertainty estimation using deep ensembles, in: *Advances in Neural Information Processing Systems*.
- [15] Lane, C.A., Hardy, J., Schott, J.M., 2018. Alzheimer’s disease. *European Journal of Neurology*.
- [16] Lee, Y.L., Tsai, Y.H., Chiu, W.C., Lee, C.Y., 2023. Multimodal prompting with missing modalities for visual recognition, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 14943–14952. doi:10.1109/CVPR52729.2023.01435.
- [17] Lepikhin, D., Lee, H., Xu, Y., Chen, D., Firat, O., Huang, Y., Krikun, M., Shazeer, N., Chen, Z., 2021. GShard: Scaling giant models with conditional computation and automatic sharding, in: *International Conference on Learning Representations*.
- [18] Lewis, M., Bhosale, S., Dettmers, T., Goyal, N., Zettlemoyer, L., 2021. BASE layers: Simplifying training of large, sparse models, in: *Proceedings of the 38th International Conference on Machine Learning*, PMLR. pp. 6265–6274.
- [19] Li, S., Chen, C., Han, J., 2025a. SimMLM: A simple framework for multi-modal learning with missing modality, in: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 24068–24077.
- [20] Li, Y., Dong, Y., Wu, Y., Yan, H., Gao, L., 2025b. Alzheimer’s disease recognition based on adaptive graph normalization flow for incomplete multimodal data fusion, in: *Medical Image Computing and Computer Assisted Intervention – MICCAI 2025*, Springer Nature Switzerland. pp. 64–73. doi:10.1007/978-3-032-04984-1_7.
- [21] Lin, R., Hu, H., 2023. MissModal: Increasing robustness to missing modality in multimodal sentiment analysis. *Transactions of the Association for Computational Linguistics* 11, 1686–1702. doi:10.1162/tac1_a_00628.
- [22] Liu, M., Cheng, D., Yan, W., Alzheimer’s Disease Neuroimaging Initiative, 2018. Classification of alzheimer’s disease by combination of convolutional and recurrent neural networks using fdg-pet images. *Frontiers in Neuroinformatics*.
- [23] Lundberg, S.M., Lee, S., 2017. A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*.
- [24] Ma, M., Ren, J., Zhao, L., Tulyakov, S., Wu, C., Peng, X., 2021. Smil: Multimodal learning with severely missing modality. *Proceedings of the AAAI Conference on Artificial Intelligence*.
- [25] Mustafa, B., Riquelme, C., Puigcerver, J., Jenatton, R., Houlsby, N., 2022. Multimodal contrastive learning with LIMoE: The language–image mixture of experts, in: *Advances in Neural Information Processing Systems*. doi:10.52202/068431-0695.
- [26] Ngiam, J., Khosla, A., Kim, M., Nam, J., Lee, H., Ng, A.Y., 2011. Multimodal deep learning, in: *Proceedings of the 28th International Conference on Machine Learning*, pp. 689–696.
- [27] Ovadia, Y., Fertig, E., Ren, J., Nado, Z., Sculley, D., Nowozin, S., Dillon, J., Lakshminarayanan, B., Snoek, J., 2019. Can you trust your model’s uncertainty? evaluating predictive uncertainty under dataset shift, in: *Advances in Neural Information Processing Systems*.
- [28] Parcalabescu, L., Frank, A., 2023. MM-SHAP: A performance-agnostic metric for measuring multimodal contributions in vision and language models and tasks, in: *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*, Association for Computational Linguistics. pp. 4032–4059. doi:10.18653/v1/2023.acl-long.223.
- [29] Park, J., Park, K., Kim, D., Park, J., Choi, J.W., 2025. Layer-wise modality decomposition for interpretable multimodal sensor fusion, in: *Advances in Neural Information Processing Systems*.
- [30] Rahman, W., Hasan, M.K., Lee, S., Zadeh, A.B., Mao, C., Morency, L.P., Hoque, E., 2020. Integrating multimodal

- information in large pretrained transformers. Proceedings of the 58th annual meeting of the association for computational linguistics .
- [31] Ribeiro, M.T., Singh, S., Guestrin, C., 2016. “why should i trust you?”: Explaining the predictions of any classifier. Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining .
- [32] Riquelme, C., Puigcerver, J., Mustafa, B., Neumann, M., Jenatton, R., Pinto, A.S., Keysers, D., Houlsby, N., 2021. Scaling vision with sparse mixture of experts, in: Advances in Neural Information Processing Systems.
- [33] Rudin, C., 2019. Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nature Machine Intelligence* .
- [34] Sadeghi, Z., Alizadehsani, R., Cifci, M.A., Kausar, S., Rehman, R., Mahanta, P., Bora, P.K., Almasri, A., Alkhawaldeh, R.S., Hussain, S., et al., 2024. A review of explainable artificial intelligence in healthcare. *Computers and Electrical Engineering* .
- [35] Scheltens, P., De Strooper, B., Kivipelto, M., Holstege, H., Chételat, G., Teunissen, C.E., Cummings, J., van der Flier, W.M., 2021. Alzheimer’s disease. *The Lancet* .
- [36] Selvaraju, R.R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., Batra, D., 2017. Grad-cam: Visual explanations from deep networks via gradient-based localization. Proceedings of the IEEE international conference on computer vision .
- [37] Shazeer, N., Mirhoseini, A., Maziarz, K., Davis, A., Le, Q., Hinton, G., Dean, J., 2017. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer, in: International Conference on Learning Representations.
- [38] Suk, H.I., Lee, S.W., Shen, D., Alzheimer’s Disease Neuroimaging Initiative, 2014. Hierarchical feature representation and multimodal fusion with deep learning for ad/mci diagnosis. *NeuroImage* .
- [39] Sundararajan, M., Taly, A., Yan, Q., 2017. Axiomatic attribution for deep networks. International Conference on Machine Learning .
- [40] Sutter, T.M., Daunhawer, I., Vogt, J.E., 2021. Generalized multimodal elbo. International Conference on Learning Representations .
- [41] Tonekaboni, S., Joshi, S., McCradden, M.D., Goldenberg, A., 2019. What clinicians want: contextualizing explainable machine learning for clinical end use. Machine learning for healthcare conference .
- [42] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, L., Polosukhin, I., 2017. Attention is all you need, in: Advances in Neural Information Processing Systems.
- [43] Venugopalan, J., Tong, L., Hassanzadeh, H.R., Wang, M.D., 2021. Multimodal deep learning models for early detection of alzheimer’s disease stage. *Scientific Reports* .
- [44] Wachter, S., Mittelstadt, B., Russell, C., 2018. Counterfactual explanations without opening the black box. *Harvard Journal of Law & Technology* .
- [45] Wang, H., Chen, Y., Ma, C., Avery, J., Hull, L., Carneiro, G., 2023a. Multi-modal learning with missing modality via shared-specific feature modelling, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 15878–15887.
- [46] Wang, Y., Cui, Z., Li, Y., 2023b. Distribution-consistent modal recovering for incomplete multimodal learning, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, pp. 22025–22034.
- [47] Weiner, M.W., Veitch, D.P., Aisen, P.S., Beckett, L.A., Cairns, N.J., Green, R.C., Harvey, D., Jack, C.R., Jagust, W., Liu, E., et al., 2013. The alzheimer’s disease neuroimaging initiative: A review of papers published since its inception. *Alzheimer’s & Dementia* .
- [48] Woo, S., Lee, S., Park, Y., Nugroho, M.A., Kim, C., 2023. Towards good practices for missing modality robust action recognition, in: Proceedings of the AAAI Conference on Artificial Intelligence, pp. 2776–2784. doi:10.1609/aaai.v37i3.25378.
- [49] Wu, M., Goodman, N.D., 2018. Multimodal generative models for scalable weakly-supervised learning. *Advances in Neural Information Processing Systems* .
- [50] Wu, R., Wang, H., Chen, H.T., Carneiro, G., 2024. Deep multimodal learning with missing modality: A survey. arXiv preprint arXiv:2409.07825 .
- [51] Xin, J., Yun, S., Peng, J., Choi, I., Ballard, J.L., Chen, T., Long, Q., 2025. I^2 MoE: Interpretable multimodal interaction-aware mixture-of-experts, in: Proceedings of the 42nd International Conference on Machine Learning, PMLR. pp. 68870–68888.
- [52] Yang, Z., Wang, H., Hu, D., 2025. Efficient quantification of multimodal interaction at sample level, in: Proceedings of the 42nd International Conference on Machine Learning, PMLR. pp. 71302–71317.
- [53] Yu, H., Qi, Z., Jang, L.K., Salakhutdinov, R., Morency, L.P., Liang, P.P., 2024. MMoE: Enhancing multimodal models with mixtures of multimodal interaction experts, in: Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing, Association for Computational Linguistics. pp. 10006–10030. doi:10.18653/v1/2024.emnlp-main.558.
- [54] Yun, S., Choi, I., Peng, J., Wu, Y., Bao, J., Zhang, Q., Xin, J., Long, Q., Chen, T., 2024. Flex-moe: Modeling arbitrary modality combination via the flexible mixture-of-experts. *Advances in Neural Information Processing Systems* .
- [55] Zhan, Y., Yang, R., You, J., Huang, M., Liu, W., Liu, X., 2025. A systematic literature review on incomplete multimodal learning: techniques and challenges. *Systems Science & Control Engineering* .
- [56] Zhang, C., Chu, X., Ma, L., Zhu, Y., Wang, Y., Wang, J., Zhao, J., 2022. M3care: Learning with missing modalities in multimodal healthcare data. Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining .
- [57] Zhang, D., Wang, Y., Zhou, L., Yuan, H., Shen, D., Initiative, A.D.N., et al., 2011. Multimodal classification of alzheimer’s disease and mild cognitive impairment. *NeuroImage* .
- [58] Zhao, J., Li, R., Jin, Q., 2021. Missing modality imagination network for emotion recognition with uncertain missing modalities, in: Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing, Association for Computational Linguistics. pp. 2608–2618. doi:10.18653/v1/2021.acl-long.203.
- [59] Zhou, R., Zhou, H., Su, Y., Chen, B.Y., Zhang, Y., He, L., Alzheimer’s Disease Neuroimaging Initiative, 2026. Adaptive clinical-aware latent diffusion for multimodal brain image generation and missing modality imputation. arXiv preprint arXiv:2603.09931 doi:10.48550/arXiv.2603.09931.
- [60] Zhou, Y., Lei, T., Liu, H., Du, N., Huang, Y., Zhao, V., Dai, A.M., Chen, Z., Le, Q.V., Laudon, J., 2022. Mixture-of-experts with expert choice routing, in: Advances in Neural Information Processing Systems. doi:10.52202/068431-0515.